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Theory of the Dielectric Dispersion due to the Interfacial Polarization and its Application to Emulsions

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With 3 figures and 4 tables

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List of Symbols

ϵ , κ , ϵ^* , κ^* , γ : the dielectric constant, the electrical conductivity, the complex dielectric constant, the complex conductivity and the imaginary part of the complex dielectric constant¹⁾ of the dispersion system respectively.

Symbols with the subscript m or p such as ϵ_m , κ_m , ϵ_m^* , κ_m^* and γ_m or ϵ_p , κ_p , ϵ_p^* , κ_p^* and γ_p , refer to the continuous medium or to the disperse phase respectively.

ω : the angular frequency.

Among these quantities there hold the following relations:

$$\epsilon^* = \frac{q}{j\omega} \kappa^* = \epsilon + \frac{q}{j\omega} \kappa = \epsilon - j\gamma, \quad [1]$$

$$\epsilon_m^* = \frac{q}{j\omega} \kappa_m^* = \epsilon_m + \frac{q}{j\omega} \kappa_m = \epsilon_m - j\gamma_m \quad [2]$$

and

$$\epsilon_p^* = \frac{q}{j\omega} \kappa_p^* = \epsilon_p + \frac{q}{j\omega} \kappa_p = \epsilon_p - j\gamma_p, \quad [3]$$

where $j = \sqrt{-1}$, numerical factor $q = 112.94 \times 10^{11}$ and the conductivities are expressed in Ω/cm units.

Φ : the volume fraction of the disperse phase.

ϵ_h , ϵ_l : the limiting values of ϵ at higher and lower frequencies respectively.

κ_h , κ_l : the limiting values of κ at higher and lower frequencies respectively.

τ : the relaxation time of a dielectric dispersion.

1. Introduction

It is well known that a heterogeneous mixture composed of two phases which differ from each other in dielectric constant and electrical conductivity shows a dielectric dispersion due to the so-called interfacial polarization²⁾.

¹⁾ The imaginary part $\frac{q}{\omega} \kappa$ of the complex dielectric constant is usually denoted by ϵ'' . In this paper a symbol γ is used instead of ϵ'' in order to avoid confusion in complicated expressions.

²⁾ This phenomenon is different from the interfacial polarization which was pointed out by Fricke and Curtis (1).

Several studies have already been carried out experimentally and theoretically on this dielectric dispersion.

A possibility of the dielectric dispersion caused by a heterogeneous structure of dielectrics was first pointed out by Maxwell (2) by means of a stratified model. This stratified model is practically of no use, since the distinction between the dispersion medium and the disperse phase is left out of consideration in this model. Wagner (3) proposed a theory of the interfacial polarization for a sparsely distributed dispersion of spherical particles, and showed a solution of his equation only for a special case of lower concentrations and the non-conductive medium. The solution given by Wagner did not agree with experimental results especially at higher concentrations as shown later. Recently Fricke (4) has extended the Wagner theory to an ellipsoidal dispersion.

On the other hand, several investigators observed dielectric dispersions due to the interfacial polarization in emulsions.

In the earlier studies Sillars (5) found $\tan \delta$ due to the interfacial polarization for emulsions of water in wax, concentrations of the internal phase being less than 7% by volume. His results did not agree with the theoretical prediction given by Wagner. Hamon (6) measured the dielectric loss factors of mixtures of spherical particles of an organic semiconductor (copper phthalocyanine) in paraffin wax, and obtained reasonably good agreement with the Wagner theory, volume concentrations of the internal phase being less than 3%. Kharadly and Jackson (7) observed dielectric dispersions due to the interfacial polarization for emulsions of polystyrene and nitrobenzene in the mixture of paraffin wax and polythene at concentrations up to 46%. Their theoretical analysis, which is similar to the Wagner theory, failed to account for the experimental results of the emulsions at concentrations above 20%. Recently Dryden and Meakins (8) have reported the dielectric absorption of the Maxwell-Wagner type for the water-in-wool wax emulsions, concentrations being less than 38%. Their results are insufficient for the rigorous consideration of the interfacial polarization, since they gave no data on the real part of the dielectric constant.

In the previous investigations (9, 10), the dielectric constants and conductivities of nujol and carbon tetrachloride-in-water

emulsions were measured for high concentrations up to 85%, and no dielectric dispersion due to the interfacial polarization could be observed over a frequency range of 20 cps. to 3 mc. The observed dielectric constants and conductivities of these O/W emulsions did not agree with the *Wiener* equation, which is a special case of the *Wagner* theory, and were expressed better by the equation of the *Bruggeman* type (11) over the whole range of concentration.

Up to the present, dielectric dispersions due to the interfacial polarization have been observed only in the water-in-oil (W/O) type emulsions as mentioned above, and no example of the interfacial polarization has been found in the oil-in-water (O/W) type yet.

It should also be noted for characteristics of dispersion systems that O/W emulsions show different dielectric constants and conductivities from those of the W/O type with the same volume fraction of the oil. In practice, these characteristics are often utilized as a crude measure of the determination of emulsion type and the detection of phase inversion (12).

Although the interfacial polarization is of practical interest especially in concentrated emulsions, it appears to us that its theoretical treatment for a spherical dispersion has not been developed beyond the *Wagner* theory.

In the present paper, the author has intended to deal with the theories on the interfacial polarization of spherical dispersions. First the validity of the *Maxwell* theory is discussed briefly. Next the general solution of the *Wagner* theory is shown, and the characteristics of this solution are discussed for emulsions. Finally a new theory of the interfacial polarization is developed and is applied to the practical cases of emulsions.

2. The Maxwell Theory

Maxwell (2) explained the dielectric dispersion due to the interfacial polarization of a heterogeneous system by means of a stratified model.

He obtained the complex dielectric constant ϵ^* and the complex conductivity κ^* for a heterogeneous system in the stratified structure as follows:

$$\epsilon^* = \epsilon_m^* \frac{\epsilon_p^*}{\epsilon_p^* + \Phi(\epsilon_m^* - \epsilon_p^*)} \quad [4]$$

and

$$\kappa^* = \kappa_m^* \frac{\kappa_p^*}{\kappa_p^* + \Phi(\kappa_m^* - \kappa_p^*)} \quad [5]$$

Then we obtain

$$\epsilon^* = \epsilon_h + \frac{\epsilon_l - \epsilon_h}{1 + j\omega\tau} + \frac{q}{j\omega} \kappa_l \quad [6]$$

and

$$\kappa^* = \kappa_l + \frac{j\omega\tau(\kappa_h - \kappa_l)}{1 + j\omega\tau} + \frac{j\omega}{q} \epsilon_h, \quad [7]$$

where

$$\epsilon_h = \epsilon_m \frac{\epsilon_p}{\epsilon_p + \Phi(\epsilon_m - \epsilon_p)}, \quad [8]$$

$$\epsilon_l = \frac{\epsilon_m \kappa_p^2 + \Phi(\epsilon_p \kappa_m^2 - \epsilon_m \kappa_p^2)}{[\kappa_p + \Phi(\kappa_m - \kappa_p)]^2}, \quad [9]$$

$$\epsilon_l - \epsilon_p = \frac{(\epsilon_m \kappa_p - \epsilon_p \kappa_m)^2 \Phi(1 - \Phi)}{[\epsilon_p + \Phi(\epsilon_m - \epsilon_p)][\kappa_p + \Phi(\kappa_m - \kappa_p)]^2}, \quad [10]$$

$$\kappa_h = \frac{\kappa_m \epsilon_p^2 + \Phi(\kappa_p \epsilon_m^2 - \kappa_m \epsilon_p^2)}{[\epsilon_p + \Phi(\epsilon_m - \epsilon_p)]^2}, \quad [11]$$

$$\kappa_l = \kappa_m \frac{\kappa_p}{\kappa_p + \Phi(\kappa_m - \kappa_p)}, \quad [12]$$

$$\kappa_h - \kappa_l = \frac{(\epsilon_m \kappa_p - \epsilon_p \kappa_m)^2 \Phi(1 - \Phi)}{[\kappa_p + \Phi(\kappa_m - \kappa_p)][\epsilon_p + \Phi(\epsilon_m - \epsilon_p)]^2}, \quad [13]$$

$$\tau = \frac{\epsilon_p + \Phi(\epsilon_m - \epsilon_p)}{\kappa_p + \Phi(\kappa_m - \kappa_p)} \cdot \frac{1}{q} \quad [14]$$

and

$$(\kappa_h - \kappa_l)\tau q = \epsilon_l - \epsilon_h. \quad [15]$$

Eq. [10] shows that a dielectric dispersion may arise when $\epsilon_m \kappa_p \neq \epsilon_p \kappa_m$. As is easily known from the stratified model, the *Maxwell* theory is unable to explain considerable changes of dielectric constants and conductivities on phase inversion of emulsions, since no distinctions are made between the continuous medium and the disperse phase of emulsions.

This theory is also of no use to explain the experimental fact that dielectric dispersions due to the interfacial polarization could not be found in O/W emulsions, while such dispersions could be observed in W/O emulsions. Moreover ϵ_h and κ_l given by eqs. [8] and [12] show no agreement with experimental values.

3. The Wagner Theory

A) The general solution of the Wagner equation

Wagner (3) showed that, for a dispersion system where spherical particles are sparsely distributed throughout the dispersion medium, the complex dielectric constant and the complex conductivity of a heterogeneous mixture are given by the following expressions:

$$\epsilon^* = \epsilon_m^* \frac{2\epsilon_m^* + \epsilon_p^* - 2\Phi(\epsilon_m^* - \epsilon_p^*)}{2\epsilon_m^* + \epsilon_p^* + \Phi(\epsilon_m^* - \epsilon_p^*)} \quad [16]$$

and

$$\kappa^* = \kappa_m^* \frac{2\kappa_m^* + \kappa_p^* - 2\Phi(\kappa_m^* - \kappa_p^*)}{2\kappa_m^* + \kappa_p^* + \Phi(\kappa_m^* - \kappa_p^*)} \quad [17]$$

Eqs. [16] and [17] have hitherto been solved only when $\Phi \ll 1$ and $\kappa_m = 0$. General solutions of these equations are derived as shown below.

Inserting eqs. [1], [2] and [3] into eq. [16], we have

$$\varepsilon^* = \left(\varepsilon_m + \frac{q}{j\omega} \kappa_m \right) \frac{[2\varepsilon_m + \varepsilon_p - 2\Phi(\varepsilon_m - \varepsilon_p)] + \frac{q}{j\omega} [2\kappa_m + \kappa_p - 2\Phi(\kappa_m - \kappa_p)]}{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)] + \frac{q}{j\omega} [2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)]} \quad [18]$$

Of we put, for simplicity,

$$\left. \begin{aligned} A &= \varepsilon_m \\ B &= q \kappa_m \\ E &= 2\varepsilon_m + \varepsilon_p - 2\Phi(\varepsilon_m - \varepsilon_p) \\ F &= q[2\kappa_m + \kappa_p - 2\Phi(\kappa_m - \kappa_p)] \\ G &= 2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p) \\ H &= q[2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)], \end{aligned} \right\} \quad [19]$$

Eq. [18] is transformed to

$$\begin{aligned} \varepsilon^* &= \left(A + \frac{B}{j\omega} \right) \frac{E + \frac{F}{j\omega}}{G + \frac{H}{j\omega}}, \quad [20] \\ &= \frac{AF + BE + j\omega AE + \frac{1}{j\omega} BF}{H + j\omega G}, \\ &= \frac{AE}{G} + \frac{\frac{AF + BE}{H} - \frac{BFG}{H^2} - \frac{AE}{G}}{1 + j\omega \frac{G}{H}} + \frac{1}{j\omega} \cdot \frac{BF}{H}. \end{aligned}$$

[21]

Hence eq. [18] is rewritten in a simplified form:

$$\varepsilon^* = \varepsilon_h + \frac{\varepsilon_l - \varepsilon_h}{1 + j\omega \tau} + \frac{q}{j\omega} \kappa_l, \quad [22]$$

where

$$\varepsilon_h = \frac{AE}{G} = \varepsilon_m \frac{2\varepsilon_m + \varepsilon_p - 2\Phi(\varepsilon_m - \varepsilon_p)}{2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)}, \quad [23]$$

$$\varepsilon_l = \frac{AF + BE}{H} - \frac{BFG}{H^2} = \frac{\varepsilon_m(2\kappa_m + \kappa_p)^2 + [(9\varepsilon_p - 2\varepsilon_m)\kappa_m^2 - 8\varepsilon_m\kappa_m\kappa_p + \varepsilon_m\kappa_p^2]\Phi - 2\varepsilon_m(\kappa_m - \kappa_p)^2\Phi^2}{[2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)]^2}, \quad [24]$$

$$\varepsilon_l - \varepsilon_h = \frac{AF + BE}{H} - \frac{BFG}{H^2} - \frac{AE}{G} = \frac{9(\varepsilon_m\kappa_p - \varepsilon_p\kappa_m)^2\Phi(1-\Phi)}{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)][2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)]^2} \quad [25]$$

and

$$\tau = \frac{G}{H} = \frac{2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)}{2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)} \cdot \frac{1}{q}. \quad [26]$$

In a similar manner we obtain

$$\kappa^* = \kappa_l + \frac{j\omega \tau (\kappa_h - \kappa_l)}{1 + j\omega \tau} + \frac{j\omega}{q} \varepsilon_h, \quad [27]$$

in which

$$\kappa_l = \frac{BF}{qH} = \kappa_m \frac{2\kappa_m + \kappa_p - 2\Phi(\kappa_m - \kappa_p)}{2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)}, \quad [28]$$

$$\kappa_h = \frac{1}{q} \left[\frac{AF + BE}{G} - \frac{AEH}{G^2} \right] = \frac{\kappa_m(2\varepsilon_m + \varepsilon_p)^2 + [(9\kappa_p - 2\kappa_m)\varepsilon_m^2 - 8\kappa_m\varepsilon_m\varepsilon_p + \kappa_m\varepsilon_p^2]\Phi - 2\kappa_m(\varepsilon_m - \varepsilon_p)^2\Phi^2}{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)]^2}, \quad [29]$$

$$\kappa_h - \kappa_l = \frac{1}{q} \left[\frac{AF + BE}{G} - \frac{AEH}{G^2} - \frac{BF}{H} \right] = \frac{9(\kappa_m\varepsilon_p - \varepsilon_p\kappa_m)^2\Phi(1-\Phi)}{[2\kappa_m + \kappa_p + \Phi(\kappa_m - \kappa_p)][2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)]^2} \quad [30]$$

and

$$(\kappa_h - \kappa_l) \tau q = \varepsilon_l - \varepsilon_h. \quad [31]$$

From eqs. [22] and [25] it is easily seen that a dielectric dispersion characterized by a single relaxation time arises when $\varepsilon_m \kappa_p \neq \varepsilon_p \kappa_m$. Therefore the Cole and Cole

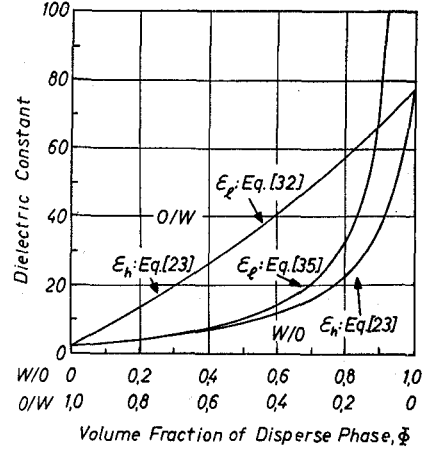


Fig. 1. Concentration dependence of dielectric constants for the O/W and W/O emulsions given by the Wagner theory

plot of eq. [22] in a complex plane is a semicircular-arc. Eq. [23] for the limiting dielectric constant ε_h at higher frequencies is the same as the Wiener equation (13) of the dielectric constant for a spherical

dispersion, and eq. [28] for κ_l is the same as the equation of the conductivity of dispersion systems used by Fricke (14).

B) Application to emulsions

We shall apply the results of the Wagner theory to the oil-water emulsions which are of practical interest. Since the conductivity of the water phase is, generally in emulsions, much larger than that of the oil phase,

eqs. [24], [28] and [29] can be simplified as follows:

i) Case $\kappa_m \gg \kappa_p$ (O/W type emulsion)

Here eqs. [24], [28] and [29] are reduced to the following respectively:

$$\varepsilon_l = \frac{2(1-\Phi)(2+\Phi)\varepsilon_m + 9\varepsilon_p\Phi}{(2+\Phi)^2}, \quad [32]$$

$$\frac{\kappa_l}{\kappa_m} = \frac{2(1-\Phi)}{2+\Phi} \quad [33]$$

and

$$\frac{\kappa_h}{\kappa_m} = \frac{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)][2\varepsilon_m + \varepsilon_p - 2\Phi(\varepsilon_m - \varepsilon_p)] - 9\varepsilon_m\varepsilon_p\Phi}{[2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)]^2}. \quad [34]$$

ii) Case $\kappa_p \gg \kappa_m$ (W/O type emulsion)

Here eqs. [24], [28] and [29] are reduced to the following respectively:

$$\varepsilon_l = \varepsilon_m \frac{1+2\Phi}{1-\Phi}, \quad [35]$$

$$\frac{\kappa_l}{\kappa_m} = \frac{1+2\Phi}{1-\Phi} \quad [36]$$

and

$$\frac{\kappa_h}{\kappa_p} = \left[\frac{3\varepsilon_m}{2\varepsilon_m + \varepsilon_p + \Phi(\varepsilon_m - \varepsilon_p)} \right]^2 \Phi. \quad [37]$$

In both cases of O/W and W/O types, ε_h can be expressed by eq. [23].

In tables 1, 2 and fig. 1 are shown the theoretical values and curves obtained respectively by eqs. [23] and [32]–[37], where the dielectric constants of the oil and the water phases are taken to be 2.50 and 76.8 (values at 30°C) respectively.

It will be noticed from tables 1, 2 and fig. 1 that both differences between ε_l and ε_h and between κ_h and κ_l are very small over the whole range of concentration for the O/W type, while these differences are very large at higher concentrations of the W/O type. Hence it is expected from the general

solution of the *Wagner* equation that the dispersions of dielectric constants and conductivities are too small to be observed in O/W emulsions and are considerable in W/O emulsions. In fact, no dielectric dispersion due to the interfacial polarization could be found in O/W emulsions (9, 10), while several workers observed the dielectric dis-

persions of this type in W/O emulsions as mentioned in the introduction.

Moreover it is explained from tables 1, 2 and fig. 1 that the considerable changes of ε_h and κ_l may occur on phase inversion of emulsions.

However it has been recognized in the previous study (9) (10) that the observed dielectric constants and conductivities of O/W emulsions do not agree with the eqs. [23] and [33] especially at higher concentrations.

Therefore it is concluded that the *Wagner* theory is not valid quantitatively for the interfacial polarization of a spherical dispersion.

4. A New Theory of the Interfacial Polarization

A) Derivation of the basic equation

The *Wagner* theory does not hold for the concentrated dispersions, since this theory was derived for a dilute dispersion system.

Now we assume that the *Wagner* relation [16] holds for the infinitesimal process such that an infinitesimally small quantity of the disperse phase is added to the dispersion

Table 1

Dielectric constants and electrical conductivities of the O/W emulsion calculated from the *Wagner* theory

Volume fraction of disperse phase	Dielectric constant		Relative conductivity	
	ε_h	ε_l	$\frac{\kappa_h}{\kappa_m}$	$\frac{\kappa_l}{\kappa_m}$
	by Eq. [23]		by Eq. [34]	
Φ	Eq. [23]	Eq. [32]	Eq. [34]	Eq. [33]
0.0	76.80	76.80	1.0000	1.0000
0.1	66.33	66.34	0.8572	0.8571
0.2	56.77	56.78	0.7274	0.7273
0.3	48.01	48.02	0.6089	0.6087
0.4	39.95	39.96	0.5002	0.5000
0.5	32.51	32.52	0.4002	0.4000
0.6	25.62	25.63	0.3078	0.3077
0.7	19.22	19.23	0.2223	0.2222
0.8	13.26	13.27	0.1429	0.1429
0.85	10.44	10.44	0.1053	0.1053
0.9	7.702	7.704	0.06900	0.06897
0.95	5.058	5.060	0.03392	0.03390
1.0	2.500	2.500	0.00000	0.00000

Table 2

Dielectric constants and electrical conductivities of the W/O emulsion calculated from the *Wagner* theory

Volume fraction of disperse phase	Dielectric constant		Relative conductivity	
	ε_h	ε_l	$\frac{\kappa_h}{\kappa_p}$	$\frac{\kappa_l}{\kappa_m}$
	by Eq. [23]		by Eq. [37]	
Φ	Eq. [23]	Eq. [35]	Eq. [37]	Eq. [36]
0.0	2.500	2.500	0.000000	1.000
0.1	3.249	3.333	0.001017	1.333
0.2	4.165	4.375	0.002511	1.750
0.3	5.309	5.714	0.004765	2.286
0.4	6.780	7.500	0.008295	3.000
0.5	8.740	10.00	0.01411	4.000
0.6	11.48	13.75	0.02436	5.500
0.7	15.59	20.00	0.04437	8.000
0.8	22.44	32.50	0.09001	13.00
0.85	27.91	45.00	0.1375	18.00
0.9	36.09	70.00	0.2271	28.00
0.95	49.70	145.0	0.4249	58.00
1.0	76.80	∞	1.0000	∞

system. Namely, for such an infinitesimally additional process, ε_m^* , ε^* and Φ in eq. [16] should be replaced as follows:

$$\varepsilon_m^* \rightarrow \varepsilon^*, \quad \varepsilon^* \rightarrow \varepsilon^* + \Delta\varepsilon^*, \quad \Phi \rightarrow \frac{\Delta\Phi'}{1-\Phi'} \quad ^{3)}$$

where $\Delta\varepsilon^*$ denotes an increment of ε^* associated with an infinitesimal addition of the disperse phase, and Φ' means the volume fraction of the disperse phase in this additional process. Therefore we obtain, for the relation between $\Delta\varepsilon^*$ and $\Delta\Phi'$,

$$\frac{2\varepsilon^* + \varepsilon_p^*}{3\varepsilon^*(\varepsilon^* - \varepsilon_p^*)} \Delta\varepsilon^* = \frac{-\Delta\Phi'}{1-\Phi'}. \quad [38]$$

This mathematical procedure is similar to that employed by *Bruggeman* (11), who derived the equation of the dielectric constant for a concentrated dispersion composed of non-conductive materials.

By succession of these infinitesimally additional processes given by eq. [38], the system will reach the final concentration Φ and dielectric constant ε^* . Hence we obtain, as a basic equation of ε^* ,

$$\int_{\varepsilon_m^*}^{\varepsilon^*} \frac{2\varepsilon^* + \varepsilon_p^*}{3\varepsilon^*(\varepsilon^* - \varepsilon_p^*)} d\varepsilon^* = \int_0^\Phi \frac{-d\Phi'}{1-\Phi'} = \log(1-\Phi). \quad [39]$$

Let us consider in fig. 2 a complex plane of ε^* , its abscissa and ordinate denoting the real part ε and the imaginary part γ respectively. A complex integral of the left side of eq. [39] is integrated along a certain contour C from ε_m^* to ε^* in the complex plane shown in fig. 2.

A contour C is determined by the correlation between ε and γ at each stage in a series of successive additions. Hence we can not know a priori an analytical representation of the contour C .

³⁾ If the volume of the dispersion medium and the disperse phase are denoted by v_m and v_p respectively, the volume fraction Φ' of the disperse phase is given by

$$\Phi' = \frac{v_p}{v_m + v_p}.$$

When a small quantity Δv_p of the disperse phase is added to the dispersion system, a quantity Φ in eq. [16] is expressed by

$$\Phi = \frac{\Delta v_p}{(v_m + v_p) + \Delta v_p}.$$

Differentiating Φ' with respect to v_p , we obtain

$$\begin{aligned} \Delta\Phi' &= \frac{\Delta v_p (v_m + v_p) - v_p \Delta v_p}{(v_m + v_p)^2} \\ &= \frac{v_m}{v_m + v_p} \cdot \frac{\Delta v_p}{v_m + v_p} = (1 - \Phi') \Phi. \end{aligned}$$

Accordingly

$$\Phi = \frac{\Delta\Phi'}{1-\Phi'}.$$

B) Evaluation of the complex integral

In a complex plane shown in fig. 2, let us consider a rectangular domain KLMN bounded by four straight lines parallel to the abscissa or the ordinate respectively through the points ε_m^* or ε^* . Since, in view of physical meanings, both ε and γ are the monotonously increasing or decreasing functions of Φ , it is obvious that the contour C is in the rectangular domain KLMN.

According to *Cauchy's* theorem (15), if a function

$$f(\varepsilon^*) = \frac{2\varepsilon^* + \varepsilon_p^*}{3\varepsilon^*(\varepsilon^* - \varepsilon_p^*)} \quad [40]$$

is regular in a domain KLMN, the integral of $f(\varepsilon^*)$ along a contour C is equal to those along the contour $\vec{KLLM}$ or $\vec{KNNM}$. The function $f(\varepsilon^*)$ given by eq. [40] has two poles, $\varepsilon^* = 0$ and $\varepsilon^* = \varepsilon_p^*$. One pole $\varepsilon^* = 0$ is the origin

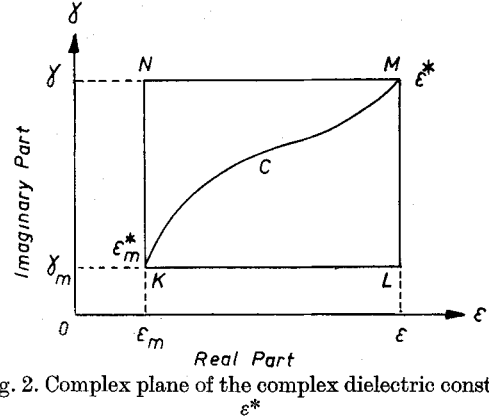


Fig. 2. Complex plane of the complex dielectric constant ε^* .

of the complex plane shown in fig. 2. Hence it is obvious that the pole $\varepsilon^* = 0$ is not in the domain KLMN. Whether another pole $\varepsilon^* = \varepsilon_p^*$ is in the domain KLMN or not, is still uncertain. If the integral of $f(\varepsilon^*)$ along the contour $\vec{KLLM}$ is equal to that along the contour $\vec{KNNM}$, then it is seen by *Morera's* theorem (15) that $f(\varepsilon^*)$ is a regular analytic function of ε^* in a domain KLMN, and a pole $\varepsilon^* = \varepsilon_p^*$ is not in this domain in consequence. The regularity of $f(\varepsilon^*)$ in the domain KLMN is proved as shown below.

The function $f(\varepsilon^*)$ is decomposed into its real part U and imaginary part V as follows.

$$f(\varepsilon^*) = -\frac{1}{3\varepsilon^*} + \frac{1}{\varepsilon^* - \varepsilon_p^*} \quad [41]$$

$$\begin{aligned} &= \left[-\frac{\varepsilon}{3(\varepsilon^2 + \gamma^2)} + \frac{\varepsilon - \varepsilon_p}{(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2} \right] \\ &\quad - j \left[\frac{\gamma}{3(\varepsilon^2 + \gamma^2)} - \frac{\gamma - \gamma_p}{(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2} \right] \\ &\equiv U - jV. \end{aligned}$$

[42]

Hence we have

$$U = -\frac{\varepsilon}{3(\varepsilon^2 + \gamma^2)} + \frac{\varepsilon - \varepsilon_p}{(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2} \quad [43]$$

and

$$V = \frac{\gamma}{3(\varepsilon^2 + \gamma^2)} - \frac{\gamma - \gamma_p}{(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2} \quad [44]$$

On the other hand, the integral of $f(\varepsilon^*)$ is decomposed into its real and imaginary parts as follows.

$$\int f(\varepsilon^*) d\varepsilon^* = \int U d\varepsilon - \int V d\gamma - j \left[\int U d\gamma + \int V d\varepsilon \right] \quad [45]$$

Each term of the integral [45] along the contour $\vec{KL}$ is given by

$$\int_{\varepsilon_m}^{\varepsilon} U d\varepsilon = \frac{1}{2} \log \frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma_m^2)^{1/3}} \quad [46]$$

$$\int_{\varepsilon_m}^{\varepsilon} V d\varepsilon = \frac{1}{3} \left[\tan^{-1} \frac{\varepsilon}{\gamma_m} - \tan^{-1} \frac{\varepsilon_m}{\gamma_m} \right] - \left[\tan^{-1} \frac{\varepsilon - \varepsilon_p}{\gamma_m - \gamma_p} - \tan^{-1} \frac{\varepsilon_m - \varepsilon_p}{\gamma_m - \gamma_p} \right] \quad [47]$$

and

$$\int V d\gamma = 0, \quad \int U d\gamma = 0. \quad [48]$$

Hence we obtain

$$\int_{\vec{KL}} f(\varepsilon^*) d\varepsilon^* = \frac{1}{2} \log \frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma_m^2)^{1/3}} + j \left[\frac{1}{3} \left(\tan^{-1} \frac{\varepsilon_m}{\gamma_m} - \tan^{-1} \frac{\varepsilon}{\gamma_m} \right) + \left(\tan^{-1} \frac{\varepsilon - \varepsilon_p}{\gamma_m - \gamma_p} - \tan^{-1} \frac{\varepsilon_m - \varepsilon_p}{\gamma_m - \gamma_p} \right) \right] \quad [49]$$

Similarly the integral [45] along the contour $\vec{LM}$ is

$$\int_{\vec{LM}} f(\varepsilon^*) d\varepsilon^* = \frac{1}{2} \log \frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2](\varepsilon^2 + \gamma_m^2)^{1/3}}{[(\varepsilon - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma^2)^{1/3}} + j \left[\frac{1}{3} \left(\tan^{-1} \frac{\gamma}{\varepsilon} - \tan^{-1} \frac{\gamma_m}{\varepsilon} \right) + \left(\tan^{-1} \frac{\gamma_m - \gamma_p}{\varepsilon - \varepsilon_p} - \tan^{-1} \frac{\gamma - \gamma_p}{\varepsilon - \varepsilon_p} \right) \right] \quad [50]$$

Combining eq. [49] with eq. [50], we obtain the integral of $f(\varepsilon^*)$ along the contour $\vec{KLLM}$ as

$$\int_{\vec{KLLM}} f(\varepsilon^*) d\varepsilon^* = \frac{1}{2} \log \frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma^2)^{1/3}} - j \left[\frac{1}{3} \tan^{-1} \frac{\varepsilon \gamma_m - \varepsilon_m \gamma}{\varepsilon \varepsilon_m + \gamma \gamma_m} - \tan^{-1} \frac{(\varepsilon - \varepsilon_p)(\gamma_m - \gamma_p) - (\varepsilon_m - \varepsilon_p)(\gamma - \gamma_p)}{(\varepsilon - \varepsilon_p)(\varepsilon_m - \varepsilon_p) + (\gamma - \gamma_p)(\gamma_m - \gamma_p)} \right] \quad [51]$$

In a similar manner the integral of $f(\varepsilon^*)$ along the contour $\vec{KNNM}$ can be evaluated. As a consequence, it is seen that the integral of $f(\varepsilon^*)$ along the contour $\vec{KLLM}$ is equal to that along the contour $\vec{KNNM}$. Hence, in view of Morera's and Cauchy's theorems, it follows that

$$\int_{\vec{KLLM}} f(\varepsilon^*) d\varepsilon^* = \int_{\vec{KNNM}} f(\varepsilon^*) d\varepsilon^* = \int_{\varepsilon_m^*(C)} f(\varepsilon^*) d\varepsilon^* \quad [52]$$

Substitution of eq. [51] in eq. [39] and separation of real and imaginary parts give

$$\frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + (\gamma_m - \gamma_p)^2](\varepsilon^2 + \gamma^2)^{1/3}} = (1 - \Phi)^2 \quad [53]$$

and

$$\tan^{-1} \frac{\varepsilon \gamma_m - \varepsilon_m \gamma}{\varepsilon \varepsilon_m + \gamma \gamma_m} = 3 \tan^{-1} \frac{(\varepsilon - \varepsilon_p)(\gamma_m - \gamma_p) - (\varepsilon_m - \varepsilon_p)(\gamma - \gamma_p)}{(\varepsilon - \varepsilon_p)(\varepsilon_m - \varepsilon_p) + (\gamma - \gamma_p)(\gamma_m - \gamma_p)} \quad [54]$$

Two eqs. [53] and [54] are the general conditions to give ε and γ or $\varkappa$.

Combining eq. [53] with eq. [54], we obtain the Bruggeman equation (11) extended to the complex numbers as

$$\frac{\varepsilon^* - \varepsilon_p^*}{\varepsilon_m^* - \varepsilon_p^*} \left(\frac{\varepsilon_m^*}{\varepsilon^*} \right)^{1/3} = 1 - \Phi \quad [55]$$

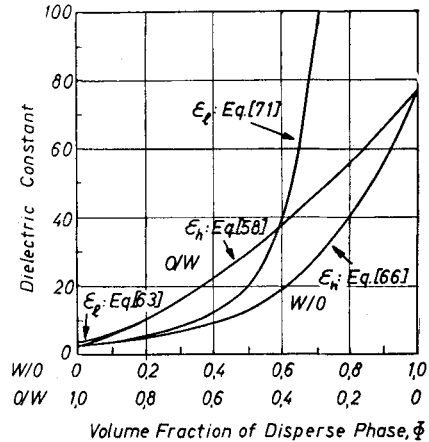


Fig. 3. Concentration dependence of dielectric constants for the O/W and W/O emulsions given by the new theory

C) Approximation for special cases

i) Case $\varkappa_m \gg \varkappa_p$ or $\gamma_m \gg \gamma_p$ (O/W type emulsion)

Here eqs. [53] and [54] lead to the following respectively:

$$\frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2](\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + \gamma_m^2](\varepsilon^2 + \gamma^2)^{1/3}} = (1 - \Phi)^2 \quad [56]$$

and

$$\tan^{-1} \frac{\varepsilon \gamma_m - \varepsilon_m \gamma}{\varepsilon \varepsilon_m + \gamma \gamma_m} = 3 \tan^{-1} \frac{(\varepsilon - \varepsilon_p)\gamma_m - (\gamma - \gamma_p)(\varepsilon_m - \varepsilon_p)}{(\varepsilon - \varepsilon_p)(\varepsilon_m - \varepsilon_p) + (\gamma - \gamma_p)\gamma_m} \quad [57]$$

a) At higher frequencies ($\omega \rightarrow \infty$)

By conditions $\varepsilon_m \gg \gamma_m$, $\varepsilon_p \gg \gamma_p$ and $\varepsilon \gg \gamma$, eq. [56] is reduced to

$$\frac{\varepsilon_h - \varepsilon_p}{\varepsilon_m - \varepsilon_p} \left(\frac{\varepsilon_m}{\varepsilon_h} \right)^{1/3} = 1 - \Phi \quad [58]$$

and eq. [57] is reduced to

$$\frac{\kappa_h}{\kappa_m} = \frac{\varepsilon_h (\varepsilon_h - \varepsilon_p) (2\varepsilon_m + \varepsilon_p)}{\varepsilon_m (\varepsilon_m - \varepsilon_p) (2\varepsilon_h + \varepsilon_p)}. \quad [59]$$

b) At lower frequencies ($\omega \rightarrow 0$)

By conditions $\gamma_m \gg \varepsilon_m$, $\gamma_p \gg \varepsilon_p$ and $\gamma \gg \varepsilon$, eqs. [56] and [57] are reduced to

$$\frac{\kappa_l - \kappa_p}{\kappa_m} \left(\frac{\kappa_m}{\kappa_l} \right)^{1/3} = 1 - \Phi \quad [60]$$

and

$$\varepsilon_l = \frac{\kappa_l [3\varepsilon_p \kappa_m + (2\varepsilon_m - 3\varepsilon_p) \kappa_l]}{\kappa_m (2\kappa_l + \varepsilon_p)}. \quad [61]$$

Especially if it is admissible to put $\kappa_l \gg \kappa_p$, eqs. [60] and [61] are reduced to

$$\frac{\kappa_l}{\kappa_m} = (1 - \Phi)^{3/2} \quad [62]$$

and

$$\frac{2\varepsilon_l - 3\varepsilon_p}{2\varepsilon_m - 3\varepsilon_p} = (1 - \Phi)^{3/2}. \quad [63]$$

ii) Case $\kappa_p \gg \kappa_m$ or $\gamma_p \gg \gamma_m$ (W/O type emulsion)

Here eqs. [53] and [54] lead to the following respectively:

$$\frac{[(\varepsilon - \varepsilon_p)^2 + (\gamma - \gamma_p)^2] (\varepsilon_m^2 + \gamma_m^2)^{1/3}}{[(\varepsilon_m - \varepsilon_p)^2 + \gamma_p^2] (\varepsilon^2 + \gamma^2)^{1/3}} = (1 - \Phi)^2 \quad [64]$$

and

$$\begin{aligned} & \frac{-i \varepsilon \gamma_m - \varepsilon_m \gamma}{\varepsilon \varepsilon_m + \gamma \gamma_m} \\ &= 3 \tan^{-1} \frac{(\varepsilon_p - \varepsilon) \gamma_p - (\gamma_p - \gamma) (\varepsilon_p - \varepsilon_m)}{(\varepsilon_p - \varepsilon) (\varepsilon_p - \varepsilon_m) + (\gamma_p - \gamma) \gamma_p}. \end{aligned} \quad [65]$$

a) At higher frequencies ($\omega \rightarrow \infty$)

By conditions $\varepsilon_m \gg \gamma_m$, $\varepsilon_p \gg \gamma_p$ and $\varepsilon \gg \gamma$, eqs. [64] and [65] are reduced to

$$\frac{\varepsilon_p - \varepsilon_h}{\varepsilon_p - \varepsilon_m} \left(\frac{\varepsilon_m}{\varepsilon_h} \right)^{1/3} = 1 - \Phi \quad [66]$$

and

$$\frac{\kappa_h}{\kappa_p} = \frac{3\varepsilon_h (\varepsilon_h - \varepsilon_m)}{(\varepsilon_p + 2\varepsilon_h) (\varepsilon_p - \varepsilon_m)}. \quad [67]$$

b) At lower frequencies ($\omega \rightarrow 0$)

By conditions $\gamma_m \gg \varepsilon_m$, $\gamma_p \gg \varepsilon_p$ and $\gamma \gg \varepsilon$, eqs. [64] and [65] are reduced to

$$\frac{\kappa_p - \kappa_l}{\kappa_p} \left(\frac{\kappa_m}{\kappa_l} \right)^{1/3} = 1 - \Phi \quad [68]$$

and

$$\varepsilon_l = \varepsilon_m \frac{\kappa_l (\kappa_p - \kappa_l)}{\kappa_m (\kappa_p + 2\kappa_l)}. \quad [69]$$

Especially if it is admissible to put $\kappa_p \gg \kappa_l$, eqs. [68] and [69] are reduced to

$$\frac{\kappa_l}{\kappa_m} = \frac{1}{(1 - \Phi)^3} \quad [70]$$

and

$$\varepsilon_l = \varepsilon_m \frac{1}{(1 - \Phi)^3}. \quad [71]$$

Eqs. [58] and [66] are the same as the Bruggeman equation (11) of the dielectric constant for a spherical dispersion.

Table 3

Dielectric constants and electrical conductivities of the O/W emulsion calculated from the new theory

Volume fraction of disperse phase Φ	Dielectric constant		Relative conductivity	
	ε_h	ε_l	$\frac{\kappa_h}{\kappa_m}$	$\frac{\kappa_l}{\kappa_m}$
	Eq. [58]	Eq. [63]	Eq. [59]	Eq. [62]
0.0	76.80	76.80	1.0000	1.0000
0.1	66.11	66.12	0.8539	0.8538
0.2	56.00	56.02	0.7158	0.7155
0.3	46.50	46.54	0.5861	0.5857
0.4	37.65	37.70	0.4654	0.4648
0.5	29.51	29.58	0.3544	0.3536
0.6	22.13	22.23	0.2541	0.2530
0.7	15.60	15.75	0.1659	0.1643
0.8	10.04	10.28	0.09175	0.08944
0.85	7.671	7.994	0.06082	0.05809
0.9	5.605	6.060	0.03473	0.03162
0.95	3.873	4.567	0.01419	0.01118
1.0	2.500	2.500	0.00000	$\sim \kappa_p/\kappa_m$

Table 4

Dielectric constants and electrical conductivities of the W/O emulsion calculated from the new theory

Volume fraction of disperse phase Φ	Dielectric constant		Relative conductivity	
	ε_h	ε_l	$\frac{\kappa_h}{\kappa_p}$	$\frac{\kappa_l}{\kappa_m}$
	Eq. [66]	Eq. [71]	Eq. [67]	Eq. [70]
0.0	2.500	2.500	0.000000	1.000
0.1	3.317	3.429	0.001312	1.372
0.2	4.499	4.883	0.004231	1.953
0.3	6.242	7.289	0.01056	2.915
0.4	8.852	11.57	0.02402	4.630
0.5	12.79	20.00	0.05190	8.000
0.6	18.69	39.06	0.1070	15.63
0.7	27.33	92.59	0.2084	37.04
0.8	39.51	312.5	0.3789	125.0
0.85	47.14	740.7	0.4966	296.3
0.9	55.87	2500.	0.6386	1000.
0.95	65.75	20000.	0.8062	8000.
1.0	76.80	0.0	1.0000	$\sim \kappa_p/\kappa_m$

In tables 3, 4 and fig. 3 are shown the theoretical values and curves obtained respectively by eqs. [58]–[63] and [66]–[71], where the dielectric constants of the oil and the water phases are taken to be 2.50 and 76.8 (values at 30°C) respectively. As is easily known from the assumptions used, eqs. [62, 63, 70] and [71] do not hold in the vicinity of $\Phi = 1$. Since the function $f(\varepsilon^*)$ is not regular at $\varepsilon^* = \varepsilon_p^*$, the results of the present theory are not valid at $\Phi = 1$. Hence, in the lowest rows ($\Phi = 1$) of tables 3 and 4 are shown the limiting values given by eqs. [60, 61, 68] and [69] at higher concentrations ($\Phi \rightarrow 1$).

D) Discussion

It should be noticed from tables 3, 4 and fig. 3 that both differences between ε_l and

ϵ_h and between κ_h and κ_i are very small over the whole range of concentration for O/W emulsions, while these differences are very large at higher concentrations of W/O emulsions. Therefore it is concluded from the present theory that the dispersions of dielectric constants and conductivities are too small to be observed in O/W emulsions and are considerable in W/O emulsions. Strictly speaking, the differences between ϵ_i and ϵ_h are rather appreciable at very high concentrations above 90%, where it may be difficult to prepare actual emulsions.

In the previous dielectric measurements (9, 10) of O/W emulsions over a wide range of frequency, no dielectric dispersion due to the interfacial polarization could be observed, and the dielectric constants and the conductivities were expressed well by eqs. [58] and [62] respectively for concentrations up to 85%. Recently Rue and Tobias (16) have observed conductivities of suspensions of glass beads in ZnBr₂ solutions, concentrations of the disperse phase being less than 40%. It was found from their results that the dependence of the conductivity of random suspensions on volume fraction is expressed better by eq. [62] rather than by eq. [33].

Author's results observed on the dielectric dispersions due to the interfacial polarization in W/O emulsions will be published in the later paper, where the validity of eqs. [66, 71, 70] and [67] would be examined.

The small differences between ϵ_i and ϵ_h for the O/W emulsion are attributable to the approximate equality between eqs. [63] and [58] for smaller values of ϵ_p . Therefore it may be suggested that, if the dielectric constant of the oil phase is much larger, the dielectric dispersion due to the interfacial polarization may be observed even in emulsions of the O/W type. Such an example of the dielectric dispersion in emulsions of the O/W type will be reported later.

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Summary

The interfacial polarization of spherical dispersions is dealt with theoretically, and is discussed for the actual cases of emulsions.

First it is shown that the *Maxwell* theory is not valid qualitatively for the dielectric properties of emulsions. Next the general solution of the *Wagner* theory is derived. This solution shows that a dielectric dispersion due to the interfacial polarization is characterized by a single relaxation time. It is found that the *Wagner* theory is not valid quantitatively for the actual emulsions.

Finally a new theory of the interfacial polarization is developed on the assumption that the *Wagner* equation holds for an infinitesimally increasing process in concentration of the dispersion system. Several dielectric properties of emulsions are interpreted quantitatively by this new theory. Dielectric dispersions due to the interfacial polarization have so far been observed only in the W/O emulsions, and no example of the interfacial polarization has been found in usual emulsions of the O/W type yet. According to the new theory it is suggested that, if the dielectric constant of the oil phase is much larger, the dielectric dispersion due to the interfacial polarization may be observed even in emulsions of the O/W type.

Zusammenfassung

Die Grenzflächenpolarisation kugelförmiger Dispersionen wird theoretisch behandelt und an Fällen von Emulsionen diskutiert.

Zuerst wird gezeigt, daß die *Maxwellsche* Theorie für die dielektrischen Eigenschaften von Emulsionen nicht einmal qualitativ gilt. Als nächstes wird eine allgemeinere Lösung der *Wagner-Theorie* abgeleitet. Diese zeigt, daß eine dielektrische Dispersion einer Grenzflächenpolarisation, durch eine einzelne Relaxationszeit charakterisiert, zuzuschreiben ist. Auch die *Wagnersche* Theorie gilt für wirkliche Emulsionen nicht quantitativ.

Schließlich wird eine neue Theorie der Grenzflächenpolarisation auf der Basis entwickelt, daß die *Wagnersche* Gleichung für ein infinitesimales Anwachsen der Konzentrationen von dispersen Systemen gilt. Zahlreiche dielektrische Eigenschaften von Emulsionen sind mit dieser neuen Theorie quantitativ interpretierbar.

Dielektrische Dispersion durch Grenzflächenpolarisation wurde bisher nur in W/O-Emulsionen beobachtet, und kein Beispiel von Grenzflächenpolarisation ließ sich in gebräuchlichen Emulsionen des O/W-Typs bis heute auffinden. Nach der neuen Theorie wird vermutet, daß die dielektrische Dispersion auf Grund von Grenzflächenpolarisation auch in Emulsionen des O/W-Typs beobachtet werden könnte, falls die Dielektrizitätskonstante der Ölphase viel größer wäre.

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The Influence of Ultrasonic Waves on the p_H , Conductivity and Gelling Time of Some Succinate Sols

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With 6 tables

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The influence of ultrasonic radiations on various colloidal systems has been investigated by several workers (1). In recent publications from this laboratory Prakash and associates (2) have made a detailed study of the influence of the ultrasonic waves on the stability of a number of sols.

In this paper our results with some gel forming succinate sols are reported. The sols investigated are (A) Zirconium Succinate, (B) Ferric Succinate and (C) Titanium Succinate.

Preparation of the Sols

(A) Zirconium succinate sol was prepared by adding sodium succinate, taken short of precipitation, to a slight excess of zirconium nitrate solution. 150 mls. of 0.357 M zirconium nitrate solution and 90 mls. of 0.25 M sodium succinate with a total volume of 990.00 mls. gave a clear sol, which was purified by dialysis for a period of 96 hrs. The sol set to transparent gels with suitable concentration of potassium chloride.

(B) Ferric succinate sol was prepared by method described by Bose and Mushran (3). Freshly precipitated and thoroughly washed ferric succinate was dissolved in 300 mls. of M/2 succinic acid by shaking in a Microid shaker. The total volume was kept 575 mls. A wine red coloured transparent sol was obtained. The sol set to transparent gels with potassium-chloride or potassium sulphate solutions of suitable concentration.

(C) Titanium succinate was prepared by adding sodium succinate, taken short of precipitation, to a slight excess of titanium tetra-chloride solution. 160 mls. of $TiCl_4$ containing 11.834 gms. of titanium per litre and 80 mls. of 0.25 M sodium succinate, with a total volume of 320 mls. gave a clear sol, which was purified by dialysis for a period of 18 hours. The sol gave transparent gels with suitable concentration of potassium chloride.

Ultrasonic Apparatus

The source of ultrasonic power was a Mullard high frequency generator with a barium titanate transducer working at 225 watts, the transducer furnishing a frequency of 1 MC/s. The transducer was immersed in a circulation bath maintained at a constant temperature. The sols were kept in jena glass vessel suitably clamped and immersed in the circulation bath. The distance between the vessel and the transducer was carefully adjusted in a manner that a good fountain was obtained.

Electrometric Measurements

The conductivity measurements were made using a L and N drum type bridge, length 1000 cms, an audio-oscillator with a tuning fork giving a constant frequency of 1000 c/sec, a tunable head phone and an immersion cell of the type GM 4221.

p_H measurements were carried out using a L and N p_H meter with a glass electrode. The instrument was properly standardised by the determination of its asymmetric potential using a standard solution of potassium hydrogen phthalate.

All electrometric measurements were made at a bath temperature of 35° C.

Gelling Time

Gelling time was determined by the method described by Bose and Mushran (4) using drops of barium sulphate suspension. The results on the influence of ultrasonic waves on the p_H and conductivity are given in tables 1-3, and in tables 4-6 are recorded the influences on gelling time.

Table 1

Showing the effect on p_H and conductivity of Zirconium Succinate Sol

Exposures	0	1	2	3	4 hrs.
p_H	3.10	3.08	3.06	3.04	3.00
Sp. Cond. $\times 10^4$	4.363	4.526	4.751	5.010	5.328 mhos.

Table 2

Showing the effect on p_H and conductivity of Ferric Succinate Sol

Exposures	0	1	2	3	4 hrs.
p_H	2.65	2.62	2.60	2.59	2.58
Sp. Cond. $\times 10^4$	2.224	2.279	2.325	2.366	2.398 mhos.

Table 3

Showing the effect on p_H and conductivity of Titanium Succinate Sol

Exposures	0	1	2	3	4 hrs.
p_H	2.55	2.50	2.45	2.41	2.38
Sp. Cond. $\times 10^3$	4.929	5.032	5.093	5.132	5.159 mhos.